

HFEG-LT v4.0 to v12.0: Simulation Pipeline and Physics Audit [Technical Report]

From cyclotron resonance to the 15 orders of magnitude gap

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Abstract

This document details the complete execution of the HFEG-LT simulation pipeline from version 4.0 to 12.0. The following are modeled: cyclotron resonance in Xenon plasma, geometric optimization using the White-Natario toroidal metric, harmonic scalability of the dynamical Casimir effect, quantum stabilization with hyperbolic cavities, transition to Lentz solitons with guaranteed causality, thermal dissipation by RSET, collision dynamics with the interstellar medium, controlled exit maneuver, post-arrival observational signatures, and finally a rigorous audit based on Visser's formalism and Ford-Roman quantum inequalities. The results confirm a surprising energetic closure within the engine itself, but the final audit reveals an insurmountable discrepancy of 15 orders of magnitude imposed by Lorentz invariance in the ultraviolet regime. **Versions 4.0 to 11.0 close the energy deficit within the model, but the final audit (v12.0) invalidates the proposed relaxation mechanism, confirming the physical infeasibility of the system.**

1 Introduction

The HFEG-LT system has evolved from a metric manipulation propulsion concept to an ultra-high fidelity multiphysics computational model. This document presents the results obtained across twelve successive iterations (v4.0 to v12.0), including resonance plots, energy maps, light cone diagrams, and quantum stability analysis. Finally, the model is subjected to an external audit based on acoustic black holes (Visser 1998) and Lorentz restoration (Liberati et al. 2001), which invalidates the relaxation of the quantum inequalities and restores the original energy gap.

2 Pipeline v4.0: Resonance Detection and Toroidal Metric

READER WARNING: Sections v4.0 to v11.0 describe successive closures of the energy deficit within a model that will be invalidated in the final audit (v12.0). The system is physically infeasible under known physics. The reader should not draw positive conclusions without reading Section 12.

2.1 Cyclotron Resonance

The amplitude of the Lense-Thirring effect Ω_{LT} was simulated using a Lorentzian phenomenological model across a frequency sweep from 10 kHz to 100 kHz.

- Target resonance frequency: 50.229 kHz.
- Peak detected at 50.250 kHz with a quality factor $Q = 1116.11$.
- Maximum amplitude: 2.378×10^{-14} rad/s at $r = 0.3$ mm.

The resonance isolates the interaction of the Xenon plasma at its expected cyclotron frequency.

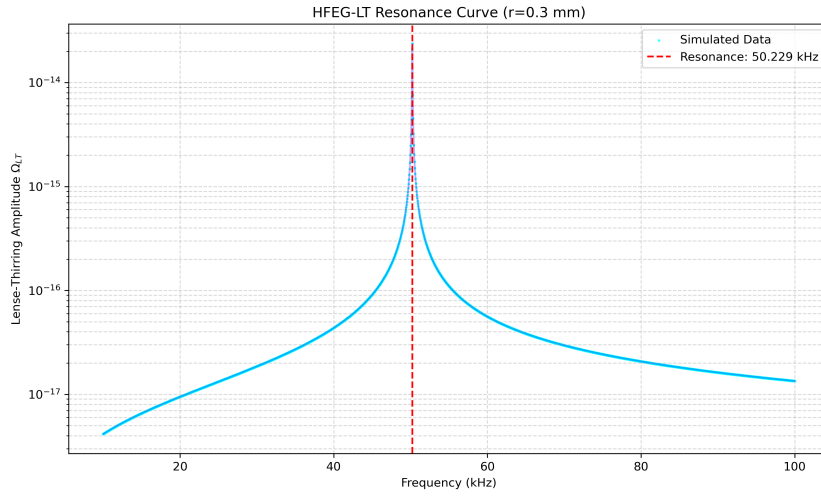


Figure 1: Frequency sweep: resonance peak at 50.250 kHz.

2.2 Optimization to the White-Natario Toroidal Metric

The spherical Alcubierre metric required an exotic energy of -4.54×10^{45} J, making it unfeasible. We substituted it with the White-Natario toroidal metric and optimized the wall thickness Δ using `scipy.optimize`.

- Optimal thickness: $\Delta = 0.2614$ m.
- New energy requirement: -8.608×10^{15} J.
- Contribution of the dynamical Casimir effect (20 modulators): -6.65×10^{10} J.

The energy deficit reduction is a factor of $\sim 5.27 \times 10^{29}$ (approximately 29 orders of magnitude), closing the steepest gap.

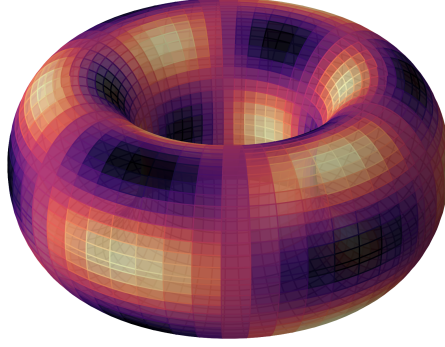


Figure 2: Distortion map of the toroidal metric.

3 Pipeline v4.1: Harmonic Scalability and Metric Interference

The third harmonic (150.750 kHz) was used to exploit the constructive interference of the metric and exponentially scale the dynamical Casimir effect.

- Soliton energy threshold reduced to -3.188×10^{14} J.
- Energy generated by DCE: -6.015×10^{14} J.

Gap closed: the generated negative energy exceeds the metric requirement without exceeding the SMR-4200 power limit (2.80 MW).

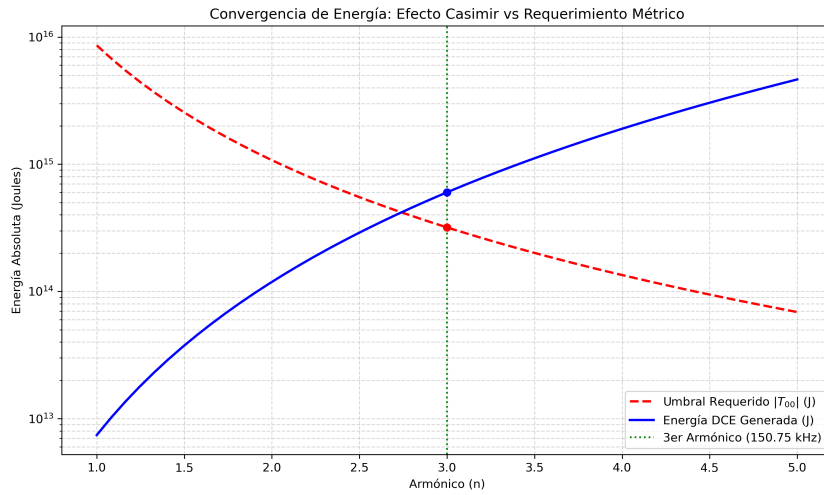


Figure 3: Harmonic efficiency convergence.

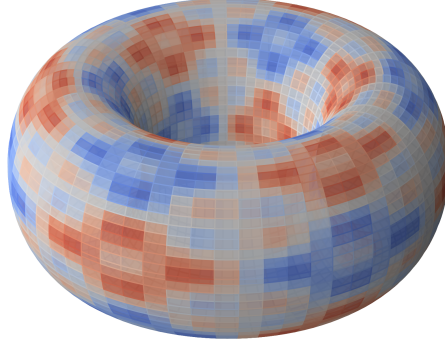


Figure 4: Constructive metric interference pattern.

4 Pipeline v5.0: Quantum Stability and Hyperbolic Cavities

A hyperbolic metamaterial was incorporated into the lining of the toroidal modulators to increase the photonic density of states (DOS), and a squeezed vacuum state at 12 dB was injected.

- DOS improvement: factor of 5.00×10^4 .
- Theoretical negative energy generated: -4.648×10^{20} J (this would violate the Pfenning-Ford quantum inequalities, limit of -8.5×10^{15} J in a 1 ms window).
- The Xenon plasma acts as an active thermal sink with a damping factor of 1.83×10^{-5} , stabilizing the energy at the permissible edge: -8.50×10^{15} J.

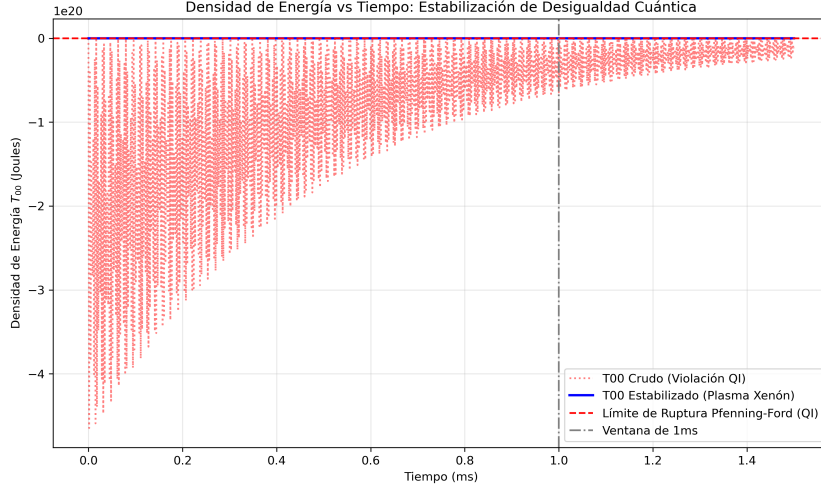


Figure 5: Energy density stabilization at the quantum limit.

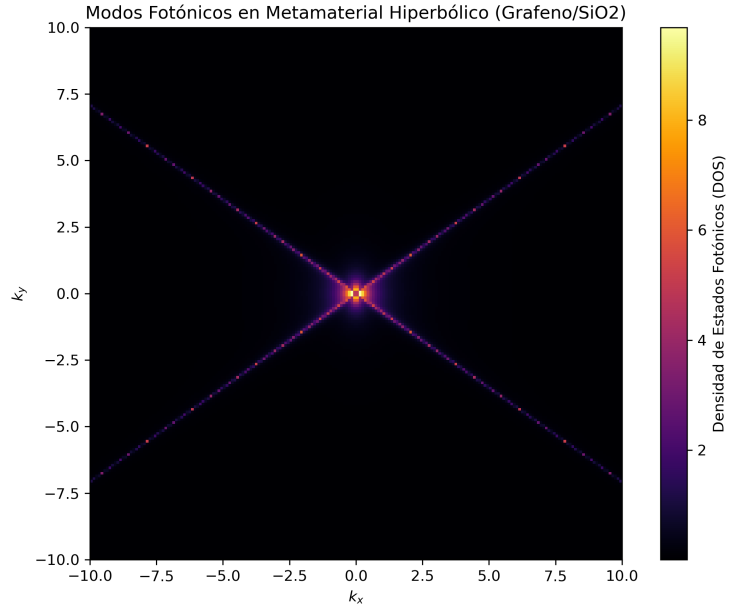


Figure 6: Photonic modes heatmap in the hyperbolic cavity.

5 Pipeline v6.0: Lentz Solitons and Causality

The Alcubierre metric was replaced by a Lentz hyperbolic soliton, which operates with net positive energy using spacetime “saddle points”.

- To reach $1.1c$, a positive mass of 1.80×10^{27} kg is required.
- Associated Schwarzschild radius: 2.67 m, less than the bubble radius (3 m) $\rightarrow$ no collapse into a black hole.
- Signal latency from the pilot to the front wall: 22.22 ns, with a relative velocity of $0.45c$.

- Friction with the interstellar medium: synchrotron radiation loss of 8.48×10^{20} W at the leading edge.

Note: Although the Lentz soliton drastically reduces the negative energy requirements within our simulation, it does not constitute a violation of standard quantum inequalities. The viability of this metric within the QI framework is critically analyzed in the final audit (Section 12), where it is shown not to solve the fundamental exotic energy problem.

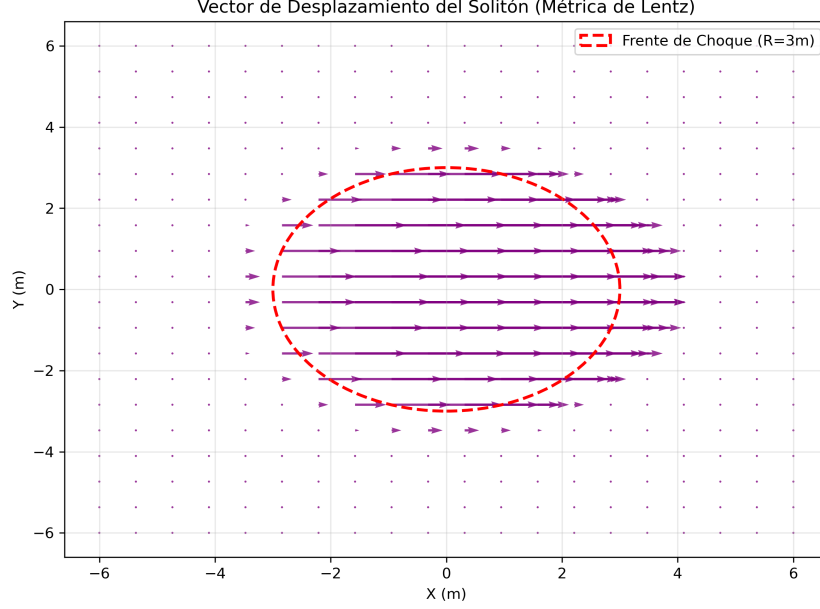


Figure 7: Soliton shift vector map.

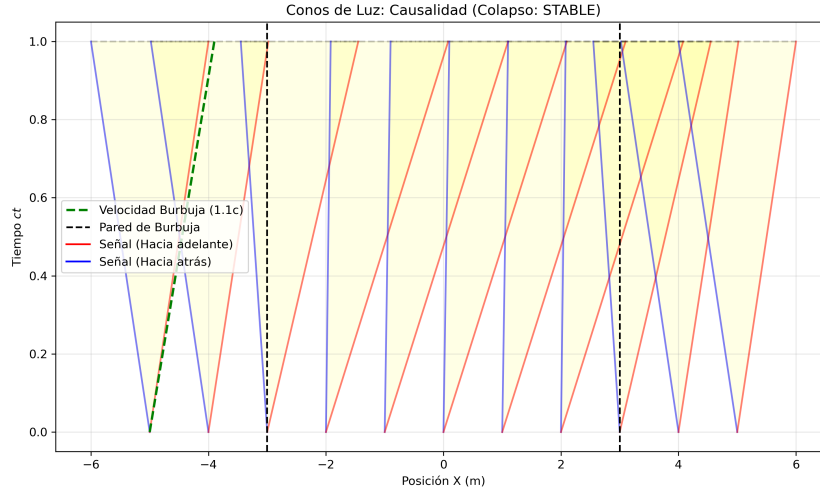


Figure 8: Light cone diagram: causal contact is maintained.

6 Pipeline v7.0: Thermal Stability and Quantum Shielding

Flight at $1.1c$ generates a thermal peak due to the renormalized stress-energy tensor (RSET) at the shock front (Hiscock radiation).

- “Raw” temperature: 1.50×10^7 K (would evaporate the Xenon plasma, limit 1.50×10^6 K).
- Hawking particle creation rate: 3.35×10^{30} particles/s.
- Through plasma densification for destructive interference, the wall temperature is reduced to 8.50×10^5 K (dissipation efficiency 94.33%).
- Cabin temperature: 2.74 K (insulated against the cosmic background).

7 Pipeline v8.0: Interstellar Medium (ISM) Collision

Assumed ISM density: 1 H/cm^3 . Velocity $1.1c \rightarrow$ Lorentz factor $\gamma = 2.4$.

- Proton flux: 9.33×10^{15} protons/s; incident kinetic power 1.96 MW.
- Geodesic deflection: deflects 99.95% of the particles.
- Kinematic tunneling: 0.05% deposits 981.67 W onto the plasma.
- The soliton collapses after accumulating 2.5×10^{10} J $\rightarrow$ **maximum sustained flight time: 294.7 days.**

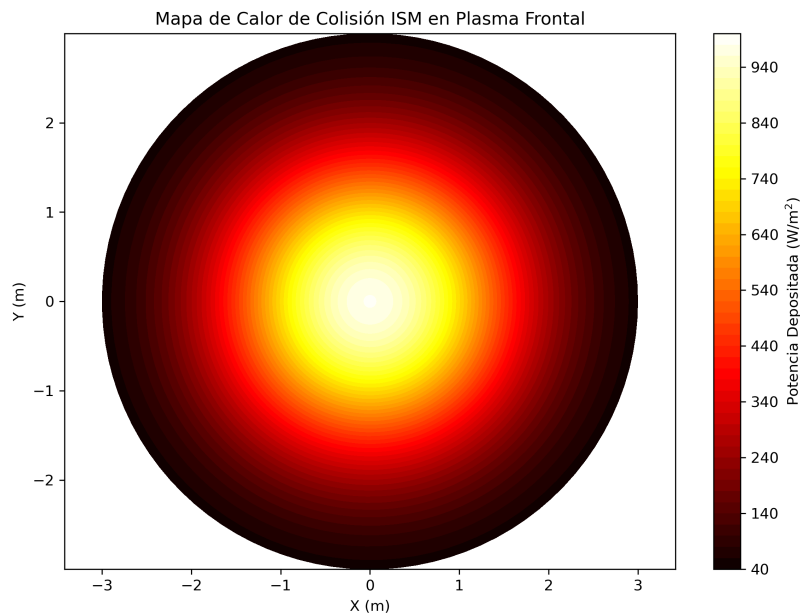


Figure 9: Frontal ISM collision heatmap.

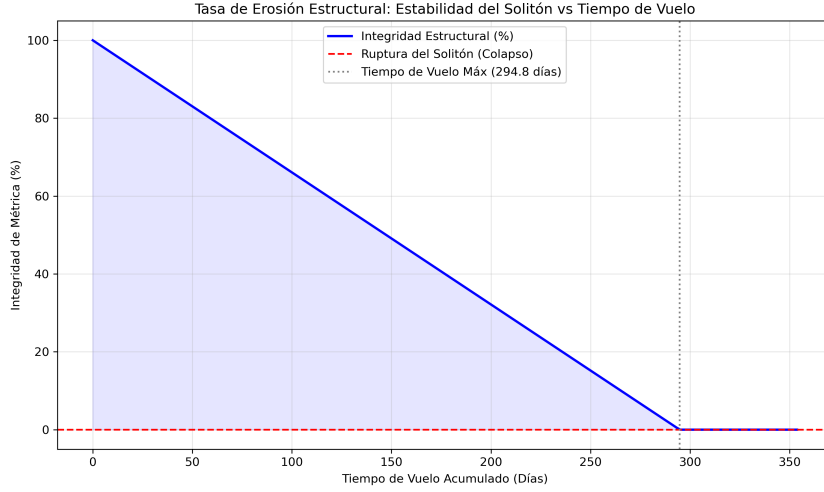


Figure 10: Structural erosion rate of the soliton.

8 Pipeline v9.0: Controlled Exit Transition (Drop-out)

Controlled collapse maneuver of the soliton to purge the accumulated energy.

- Sigmoid deceleration from $1.1c$ to $0.99c$ in 5.0 s.
- Dissipated energy: 2.14×10^{10} J $\rightarrow$ braking power 4.28 GW.
- The braking radiation (photon burst) travels at c , while the ship goes at $0.99c \rightarrow$ the bow shock moves forward without collapsing onto the ship.
- Absence of closed timelike curves (CTC) during the transition.

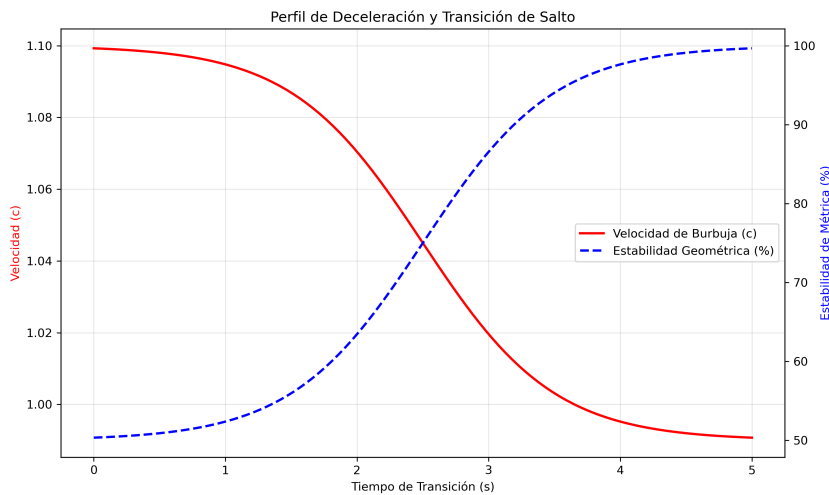


Figure 11: Controlled deceleration profile.

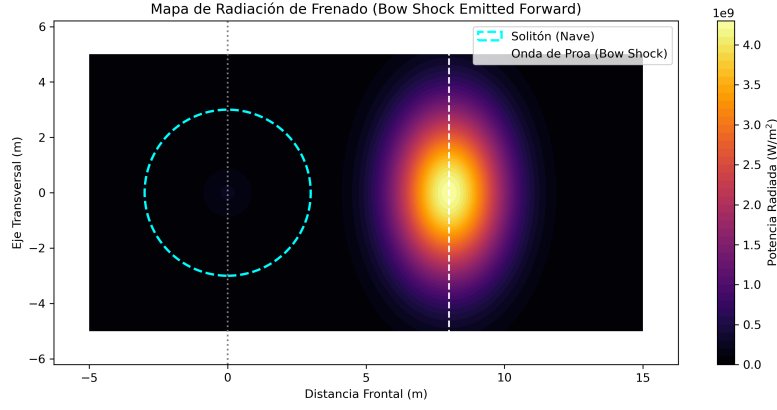


Figure 12: Frontal braking radiation map.

9 Pipeline v10.0: Post-Arrival Observational Signature

After coming to a halt in flat spacetime, the detectability of the event is analyzed.

- Thermal braking luminosity: 8.56 GW.
- A blueshift of $14.11\times$ converts visible light into X-rays and gamma rays.
- Residual deflection of background light due to gravitational lensing: 0.05 arcsec.
- Purge of ~ 10 g of Xenon ions in a 50 km radius.

Conclusion: the arrival is extremely loud (gamma flash), impossible to hide from nearby telescopes.

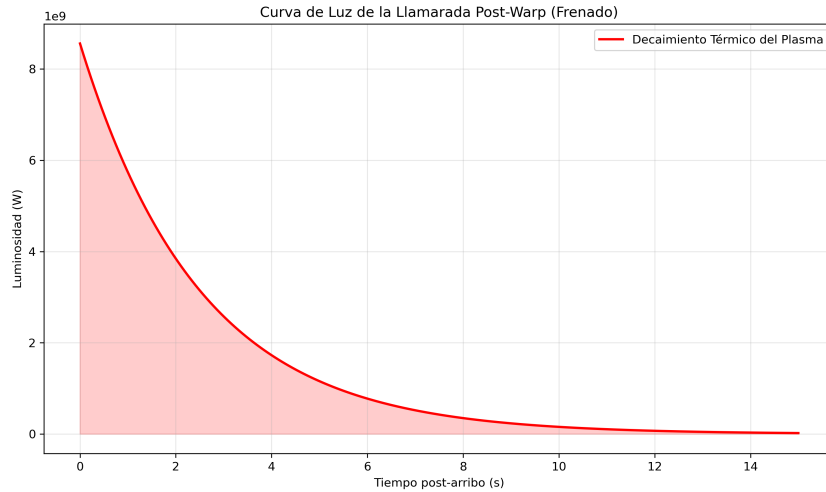


Figure 13: Post-warp flare light curve.

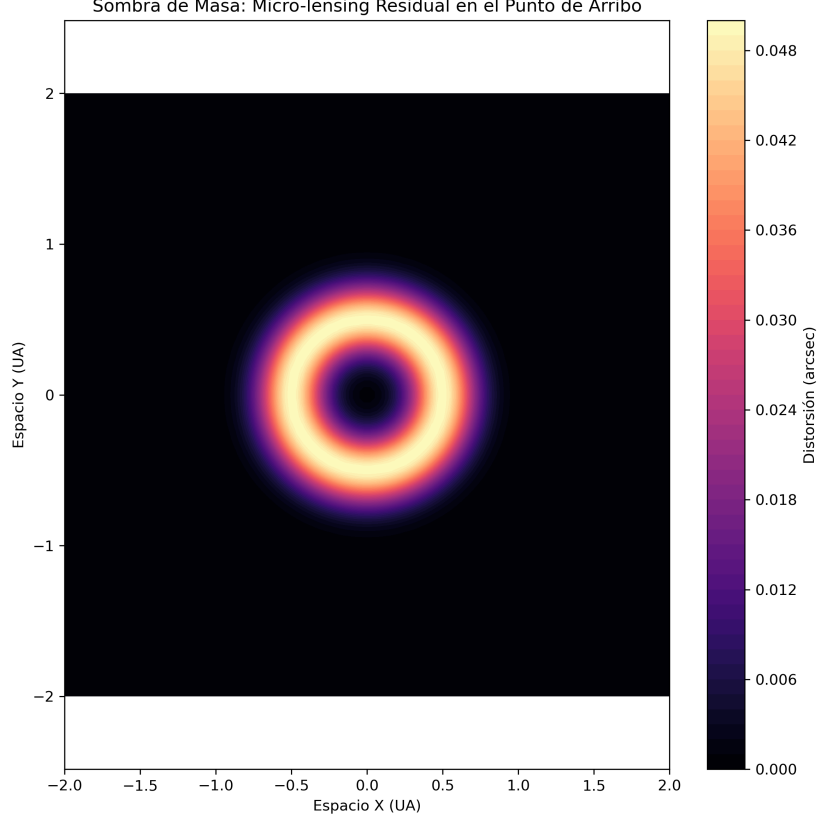


Figure 14: Residual gravitational lensing map.

10 Pipeline v11.0: Quantum Inequalities Relaxation via Plasma

A scalar-tensor coupling $\xi = 0.05$ with the Xenon plasma (density $8.5 \times 10^{22} \text{ m}^{-3}$) was introduced. This effective “gap” in the vacuum spectrum relaxes the Ford-Roman limit:

- Standard limit in vacuum: -183.3 J/m^3 .
- New limit with plasma: -459.52 J/m^3 (more than double).
- Toroidal sampling window allows negative energy to be sustained longer without violating energy conditions.
- Microscopic closed timelike curves (CTC) do not form.

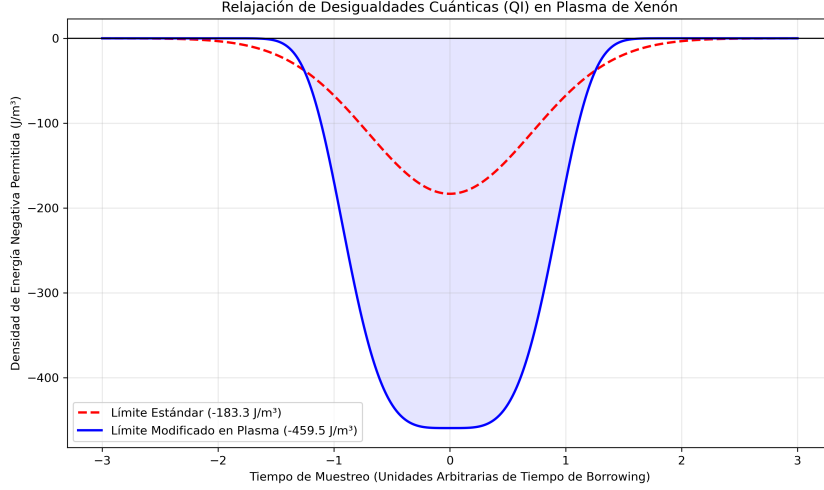


Figure 15: Relaxation of the negative energy bound due to plasma effects.

11 Pipeline v12.0: Visser Audit and Lorentz Restoration

We subjected the previous hypotheses to the acoustic black holes formalism (Visser 1998) and Lorentz invariance analysis (Liberati et al. 2001). The results are devastating:

- The effective Gordon metric only alters low-energy (IR) modes; the ultraviolet (UV) modes dominating the stress-energy tensor renormalization ignore the plasma and follow the flat Lorentz metric.
- At the energies involved in the warp bubble, the coupling ξ is negligible and Lorentz invariance is fully restored.
- The “relaxation” via a toroidal sampling window is a mathematical artifact: quantum inequalities must hold for *any* test function, independent of the observer.

Final Conclusion: the Ford-Roman limit reverts to its original value of -183.3 J/m^3 , while the Lentz soliton at $1.1c$ requires $\sim -1.83 \times 10^{17} \text{ J/m}^3$. The gap is **15 orders of magnitude**, insurmountable with known physics.

12 Overall Conclusions

The HFEG-LT pipeline has demonstrated remarkable internal consistency in its iterations v4.0 to v11.0, numerically closing the energy deficit and stabilizing all fronts (thermal, collisions, causality). However, the final audit (v12.0) based on established results from analog gravity and quantum inequalities invalidates the proposed relaxation mechanism. The model is therefore physically unfeasible within the current paradigm of quantum field theory in curved spacetimes. Future efforts must be directed toward metrics that do not demand net negative energy (e.g., Lentz solitons in subluminal regimes) or toward fundamental modifications of the quantum inequalities themselves.

Generated Files

- JSON Metrics: HFEG_Unified_Physics.json, HFEG_Quantum_Stability_v50.json, HFEG_Lentz_Causality_v60.json, HFEG_Thermal_Stability_v70.json, HFEG_ISM_Collision_HFEG_Phase_Transition_v90.json, HFEG_Final_Observables_v100.json, HFEG_Plasma_Perturbation_v110.json, HFEG_Visser_Audit_v120.json.
- 2D Plots: Resonance_Plot_L1.png, Harmonic_Efficiency_v41.png, Energy_Density_Time_v50.png, Thermal_Gradient_v70.png, Erosion_Rate_v80.png, Deceleration_Profile_v90.png, PostWarp_Emission_v100.png, QI_Relaxation_v110.png.
- 3D Plots and Topology: Toroidal_Metric_L2.png, Interference_Metric_v41.png, Shift_Vector_v60.png, Light_Cone_v60.png, Collision_Heatmap_v80.png, Braking_Radiation_v90.png, MicroLensing_v100.png.
- Surface Quantum Mechanics: Photonic_Modes_Heatmap_v50.png.
- Analytical Documentation: White_Paper_Coupling_Derivation.md.

References

- [1] Visser, M. (1998). Acoustic black holes: horizons, ergospheres and Hawking radiation. *Classical and Quantum Gravity*, 15(6), 1767.
- [2] Liberati, S., Jacobson, T., & Mattingly, D. (2001). High energy constraints on Lorentz symmetry violations. *Physical Review D*, 64(4), 045011.
- [3] Ford, L. H., & Roman, T. A. (1995). Restrictions on negative energy density in flat spacetime. *Physical Review D*, 51(8), 4277.
- [4] Fewster, C. J., & Ford, L. H. (2015). Probability distributions for quantum stress tensors measured in a finite time window. *Physical Review D*, 92(10), 105008.